

Process	AT	BT	Priority
P ₁	0	1	5
P ₂	0	2	7
P ₃	2	1	3
P ₄	5	3	2
P ₅	10	5	6

2. Apply all the CPU scheduling Algorithms for insert

Process	AT	BT	Priority
P ₁	0	4	3
P ₂	0	9	1
P ₃	0	6	2
P ₄	0	4	3
P ₅	0	9	1
P ₆	0	6	2
P ₇	1	4	3
P ₈	1	9	1
P ₉	2	6	2

3. Define following terms

starvation, convoy effect, context switching, Aging

4. mobile operating system has following task draw a table 5 rows and 3 columns

Task	AT	BT
music player	0	20
Touch event	3	1
whatsapp notification	5	2
camera launch	6	4

5. solve the following problem with all the CPU scheduling algorithms

	AT	BT
P_1	0	5
P_2	0	4
P_3	0	3

is there is context to switch in CPU algorithm then there is context which time 0.5 secs.

(i) which scheduling algorithm is most suitable for following

(ii) Draw the grad chart of SJF

(iii) why the selecting algorithm for D, improves user experience

(iv) which task receives the fastest response

Solve the following problem with round robin algorithm using quantum time of 2 and context switch time of 0.5 draw a table with 4 rows and 3 columns.

Week 2:-

Answer:

Process	AT	BT	priority
P ₁	0	4	3
P ₂	0	9	1
P ₃	0	6	2
P ₄	0	4	3
P ₅	0	9	1
P ₆	0	6	2
P ₇	1	4	3
P ₈	1	9	1
P ₉	2	6	2

FCFS:- (First come First serve)

RB: 0: [P₁ | P₂ | P₃ | P₄ | P₅ | P₆]

1: [P₁ | P₂ | P₃ | P₄ | P₅ | P₆ | P₇ | P₈]

2: [P₁ | P₂ | P₃ | P₄ | P₅ | P₆ | P₇ | P₈ | P₉]

3: [~~P₁~~ | P₂ | P₃ | P₄ | P₅ | P₆ | P₇ | P₈ | P₉]

4: [~~P₁~~ | ~~P₂~~ | P₃ | P₄ | P₅ | P₆ | P₇ | P₈ | P₉]

5: [~~P₁~~ | ~~P₂~~ | ~~P₃~~ | P₄ | P₅ | P₆ | P₇ | P₈ | P₉]

23:-

P₁	P ₅	P ₆	P ₇	P ₈	P ₉
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32:-

P₁	P ₆	P ₇	P ₈	P ₉
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38:-

P₁	P ₇	P ₈	P ₉
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42:-

P₁	P ₈	P ₉
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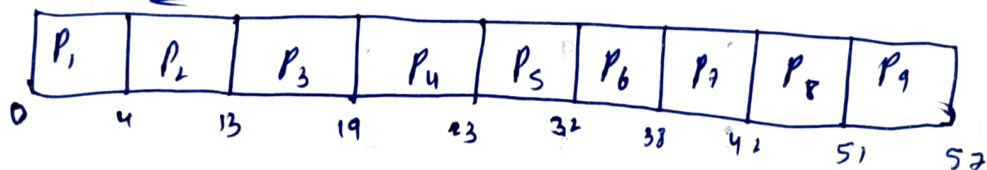
51:-

P₁	P ₉
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57:-

P₁

Gantt chart:



Process	AT	BT	LT	TAT	WT	RT
P ₁	0	4				
P ₂	0	9				
P ₃	0	6				
P ₄	0	4				
P ₅	0	9				
P ₆	0	6				
P ₇	1	4				
P ₈	1	9				
P ₉	2	6				

Week -3

Answer

Starvation:- is a situation where a process waits for a

very long time to get the required cpu or other resources because other process keep getting priority.

Ex:- ~~At low priority~~ A low-priority process may have keep waiting while high-priority process are continuously executed.

Convoy effect:- The effect occurs when a long-running process holds the cpu, causing many short process to wait behind it

Ex:- in FCFS scheduling if a long process comes first all the smaller process must wait like vehicles stuck behind a slow truck.

Context Switching:- is the process of saving the state of a currently running process and loading the saved state of another process so that the cpu can switch between them

Ex:- CPU switches from process $P_1 \rightarrow$ process P_2

Aging:- is a technique used to prevent starvation by gradually increasing the priority of a process that has been waiting for a long time.

Ex:- A process waiting for a long time gets its priority increased until it gets cpu time.

Term	meaning
starvation	process waits for too long
convoy effect	short process stuck behind a long process
Context switching	cpu switches from the one process to another
aging	increased priority of waiting process.

Week -4

Answer

i) Satisfiable scheduling algorithm:

(SRTF) (Shortest remaining Time First) is satisfiable because it gives cpu to the task with shortest remaining time. So small tasks finish quickly.

ii) Gantt chart

0 3 4 5 7 11 26

|M| T | M | W | C | | M |

M = music player

T = Touch event

W = Whatsapp notification

C = camera launch

music player $\rightarrow$ Touch event $\rightarrow$ whatsapp $\rightarrow$
camera $\rightarrow$ music

M $\rightarrow$ T $\rightarrow$ W $\rightarrow$ C $\rightarrow$ M

iii) why does it improve user experience?

SRTF give a quick response to small tasks like touch events and notification so, the mobile feels fast and responsive

iv) Touch event gets the fastest response it arrives at time 3 and starts at time 3.

response time = $3 - 3 = 0$

touch event = 0 units.

Week -1

Answer

Given that

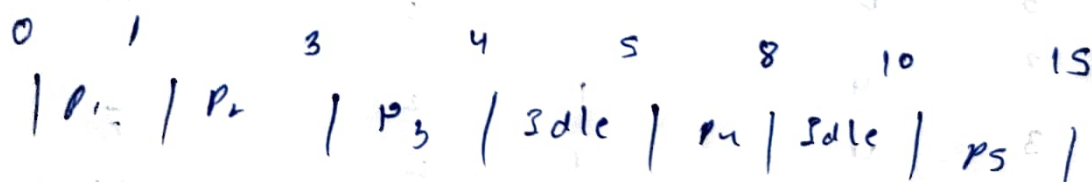
Process	AT	BT	priority
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P ₂	0	2	2
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Smaller priority number = higher priority

→ Scheduling

- at 0 P₁ and P₂ available — P₁ has higher priority → P₁
- at 1 P₂ is only available — P₂
- at 3 P₃ is available — P₃
- at 4 no process is available → CPU is idle
- at 5 P₄ arrives — P₄
- at 8 no process is available — CPU is idle
- at 10 P₅ arrives — P₅

→ Gantt chart



→ Final order

P₁ — P₂ — P₃ — P₄ — P₅